



Artwork by Paul Jackson Pollock

Parsing CSVs and the Pollock Benchmark

Polluted CSVs - Examples

Missing delimiter

```
id,name,city
1,Alice,Nuremberg
2,Bob Berlin
3,Charlie,Munich
```

Non-standard delimiter

```
id|name|city
1|Alice|Nuremberg
2|Bob|Berlin
3|Charlie|Munich
```

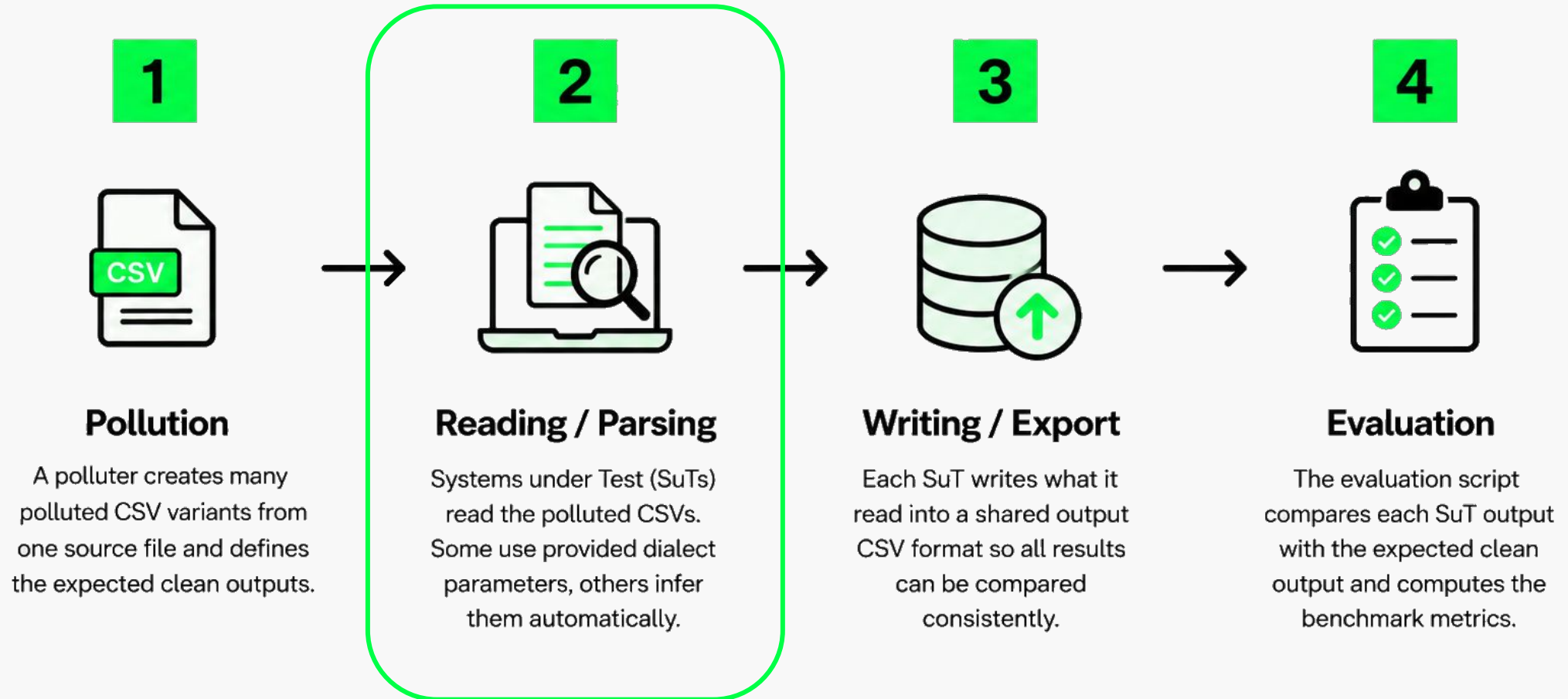
Unescaped quote character

```
id,product,description
1,Laptop,"Lightweight 14 inch model"
2,Chair,"Bob's "ergonomic" office chair"
3,Desk,"Height adjustable"
```

Multi-line header

```
product,product,sale
id,type,date
P001,book,2024-01-01
P002,laptop,2024-01-02
P003,chair,2024-01-03
```

Benchmark Overview



SUT	pollock_simple	pollock_weighted	Uses dialect info?	Runtime
custom	10.0 (soon)	10.0 (soon)	No	Python or Docker
duckdbparse	9.961516	9.599662	Yes	Python
mariadb	9.953843	9.610157	Yes	Docker
mysql	9.953843	9.610157	Yes	Docker
univocity	9.939419	7.936767	No	Docker
sqlite	9.936568	9.589233	Yes	Docker
spreaddesktop	9.929668	9.597198	N/A	N/A
libreoffice	9.925582	7.833335	Yes	Docker
pandas	9.884786	7.909017	Yes	Python
pycsv	9.724189	9.436467	No	Python
spreadweb	9.721757	9.431587	N/A	N/A
duckdbauto	9.646808	8.996221	No	Python
clevercs	9.193083	9.453858	No	Python
dataviz	5.003541	5.152075	N/A	N/A
hypoparsr	3.877452	4.400585	No	Docker
postgres	0.141977	7.872715	Yes	Docker

pollock_simple:

```
Score = mean(success)
+ mean(header_precision) + mean(header_recall) + mean(header_f1)
+ mean(record_precision) + mean(record_recall) + mean(record_f1)
+ mean(cell_precision)   + mean(cell_recall)   + mean(cell_f1)
```

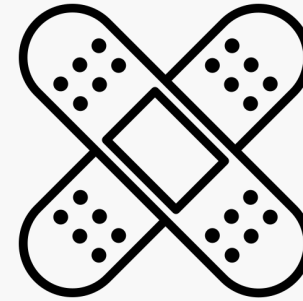
pollock_weighted: Errors weighted according to frequency of pollution type found in survey

Automatic Detection Approaches



Sniffer

e.g. maximize the number of lines + columns that can be read into a schema based on sample. Sometimes just casts everything to String...



Repair-approaches

First identify, then try to repair malformed rows. E.g. through edit distance (Tashee). Could be build upon a standard sniffer

Quirks of the Benchmark

Debatable ground truth

Some “correct” outputs are conventions, e.g. how to handle multiple headers being present. Cut off all but the first one? concatenate columnwise?

Exact string matching

Final exported CSVs are compared very strictly. Make sure your export is in the correct format expected by the benchmark.

Example:

00:00 ≠ 00:00:00

YOU PROBABLY DONT NEED TO RUN OTHER SUTs

but if you do...

Python SuTs: only pip installs (e.g. CleverCSV, DuckDB, pycsv)

Other SuTs: require Docker, often large images, sometimes very slow. (e.g. LibreOffice Calc, HypoParsr)

Reproducibility note

Some systems from the paper are not included by name because their licenses/terms did not allow public benchmarking results. They are also not in the repository for the same reason.

Struggles of current SuTs - INCONSISTENCY + Dumb things

Inconsistency

Missing delimiter

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Unescaped quote character

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Dumb things

e.g. SPACE (0x20) as delimiter

```
id name city motivation
1 Alice Nuremberg Wants to talk to B.
2 Bob Berlin Wants to talk to A.
3 Eve Munich Wants to eavesdrop on A. + B.'s conversation
```

...in conjunction with an unquoted String